

Physiology Practice Questions: DM Module

Lecture Title: Saliva and Gastric Secretions (L24)

Note: If you need any clarification, please email to: ASK-PHY@squ.edu or email to the faculty who delivered this lecture.

1. A 31-year-old man comes to the physician because of a reducible swelling in his left loin. He is diagnosed with left-sided indirect inguinal hernia. He is posted for elective surgery under general anesthesia. He receives an intramuscular injection of atropine as a preanesthetic medication. Which of the following physiological changes is most likely after this injection?

- A. Reduced sense of smell
- B. Increased taste sensation
- C. Reduced salivary secretion
- D. Increased bronchial secretion
- E. Bradycardia

2. A 54-year-old man comes to the physician because of a 5-month history of heartburn and epigastric discomfort. The symptoms are worse in night and when he postpones his meals but reduce after eating. Laboratory studies of gastric juice shows increased acid secretion in the stomach. Which of the following mechanisms best explains the increased H⁺ concentration in this patient's gastric juice?

- A. Hypertonicity of the oxyntic fluid
- B. Increased volume of the oxyntic fluid
- C. Stimulation of oxyntic cells
- D. Stimulation of chief cells
- E. Inhibition of G cells

3. A 43-year-old man comes to the physician because of a 3-week history of increased salivation. He takes some medication for his mental illness and has recently changed to a new medication. Physical examination shows drooling. Concentration of which of the following substances is most likely reduced in this patient's saliva?

- A. Sodium
- B. Chloride
- C. Bicarbonate
- D. Potassium
- E. Water

4. A 38-year-old woman comes to the physician because of a 6-month history of abdominal pain. She says that the pain reduces in intensity by eating. She smokes 1 pack of cigarettes and consumes 250 mL of wine daily. Upper GI endoscopy shows two small ulcers in the body of the stomach. She is diagnosed with peptic ulcer disease because of an increased acid secretion. Blockade of which of the following most likely reduces acid secretion in this patient?

- A. Secretin secretion
- B. Histamine H₂ receptors
- C. Pepsinogen secretion
- D. Somatostatin secretion
- E. CCK secretion

5. A 26-year-old woman comes to the office because of a 2-month history of burning pain below the sternum. He says that the pain begins in the epigastric area one hour after eating and is associated with nausea. He has had some relief from the proton pump inhibitor that was prescribed 2 weeks ago. Physical examination shows no abnormalities. Laboratory studies show an extremely high serum concentration of gastrin. Imaging studies show a mass in the tail of the pancreas. He is diagnosed with a gastrin-secreting tumor. Which of the following mechanisms best explains this patient's symptoms?

- A. Reduced gastric motility
- B. Stimulation of the parietal cells
- C. Lack of blood flow to the stomach
- D. Failure of relaxation of the lower esophageal sphincter
- E. Gall bladder contraction

6. A 49-year-old man comes to the clinic because of a 10-month history of intermittent abdominal pain. Physical examination shows epigastric tenderness to palpation. Gastric juice study shows increased acid secretion. Stool and blood studies show no abnormalities. An upper GI endoscopy shows an ulcer near the gastric antrum. He is diagnosed with peptic ulcer disease and prescribed a drug that inhibits acid secretion from the parietal cells of the stomach. Which of the following mechanisms in the parietal cells is most likely blocked by this drug?

- A. Hydrogen-potassium ATPase
- B. Acetylcholine receptors
- C. Bicarbonate-chloride exchanger
- D. Carbonic anhydrase
- E. Histamine receptors

7. A 38-year-old woman comes to the physician for a follow-up examination. She has a 2-year history of autoimmune disease of the salivary gland. Physical examination shows

dry mouth and scanty salivary secretion. Histological examination of a biopsy of the salivary gland shows atrophy of the gland. Which of the following gastrointestinal functions is most likely impaired in this patient?

- A. Digestion of proteins in mouth
- B. Reabsorption of vitamin B12
- C. Carbohydrate digestion in mouth
- D. Carbohydrate digestion in small intestine
- E. Digestion of fat in small intestine

8. A 61-year-old man comes to the clinic because of a 4-month history of burning epigastric pain. The pain is relieved with meals. He has no history of nausea or vomiting. An acid secretion stimulation test shows a gastric juice pH of 0.94 (Normal: 1.6 to 3.5). Secretion of which of the following substances most likely inhibits acid secretion in this patient?

- A. Gastrin
- B. Acetylcholine
- C. Histamine
- D. Gastrin-releasing peptide
- E. Secretin

9. A 32-year-old woman comes to the physician because of an 8-month history of progressive fatigue. In the beginning, she felt more tired than usual at the end of the day but now has difficulty walking up one flight of stairs. She also has numbness and tingling in her feet. Her menses occur at regular 28-day cycle. Her vital signs are within normal limits. Physical examination shows pale tongue and sclera, and ulcers at the angle of the mouth. Sensory examination shows diminished proprioception and vibration sensation in the lower extremities. A peripheral blood smear shows macrocytic hypochromic red blood cells and hypersegmented neutrophils. Histologic examination of a gastric mucosal biopsy shows loss of gastric glands. She is diagnosed with vitamin B12 deficiency anemia secondary to stomach disorder. Reduced secretion of which of the following substances from the stomach is the most likely cause of this patient's presentation?

- A. Hydrochloric acid
- B. Mucin
- C. Gastrin
- D. Intrinsic factor
- E. Pepsinogen
- F. Lipase

10. A 24-year-old woman comes to the physician for a routine examination. She has no complaints but gets increased salivary secretion for certain food stuff. She loves spicy

food from childhood. She says she starts salivating when she smells and sees spicy food. Which of the following mechanisms is responsible for increased salivation in this patient?

- A. Stimulation of the sympathetic nerves to the salivary glands
- B. Increased blood flow to the salivary glands
- C. Conditioned reflex
- D. Stimulation of the enteric nervous system
- E. Vagovagal reflex

Answer Key and Explanation

1. A 31-year-old man comes to the physician because of a reducible swelling in his left loin. He is diagnosed with left-sided indirect inguinal hernia. He is posted for elective surgery under general anesthesia. He receives an intramuscular injection of atropine as a preanesthetic medication. Which of the following physiological changes is most likely after this injection?

- A. Reduced sense of smell
- B. Increased taste sensation
- C. Reduced salivary secretion
- D. Increased bronchial secretion
- E. Bradycardia

Answer Key: C

Rationale: Atropine is a muscarinic receptor blocker that blocks the action of acetylcholine. Salivary and bronchial secretions are predominantly stimulated by parasympathetic nerves via acetylcholine through muscarinic receptors. Injection of this drug reduces these secretions and helps anesthetist for better intubation and is used as a preanesthetic medication before surgery. Stimulation of the parasympathetic nerves to the salivary glands is the predominant regulator of salivary secretion.

2. A 54-year-old man comes to the physician because of a 5-month history of heartburn and epigastric discomfort. The symptoms are worse in night and when he postpones his meals but reduce after eating. Laboratory studies of gastric juice shows increased acid secretion in the stomach. Which of the following mechanisms best explains the increased H⁺ concentration in this patient's gastric juice?

- A. Hypertonicity of the oxyntic fluid
- B. Increased volume of the oxyntic fluid
- C. Stimulation of oxyntic cells
- D. Stimulation of chief cells
- E. Inhibition of G cells

Answer Key: C

Rationale: Oxyntic fluid is another term for the acidic component (H^+ concentration) of gastric secretion because oxyntic (parietal) cells secrete H^+ . When the oxyntic component of gastric secretions increases, it becomes more acidic (contains more H^+). Tonicity remains same as plasma (isotonic) even with the increased activity of the oxyntic cells.

3. A 43-year-old man comes to the physician because of a 3-week history of increased salivation. He takes some medication for his mental illness and has recently changed to a new medication. Physical examination shows drooling. Concentration of which of the following substances is most likely reduced in this patient's saliva?

- A. Sodium
- B. Chloride
- C. Bicarbonate
- D. Potassium
- E. Water

Answer Key: D

Rationale: The rate of salivary secretion is increased in this patient. Composition of primary saliva alters when it passes through the ducts. The ducts remove $NaCl$ and secrete K^+ and HCO_3^- . The net effect is a hypotonic modification of the salivary fluid because more $NaCl$ is removed than ions secreted. Under slow rate of secretion, the fluid remains longer in the duct, therefore, the end saliva is hypotonic, contains less sodium and chloride and more potassium. When the secretion rate is high (as in this patient), the sodium and chloride content would be high (and potassium would be low. There is less time for potassium secretion in the duct.

4. A 38-year-old woman comes to the physician because of a 6-month history of abdominal pain. She says that the pain reduces in intensity by eating. She smokes 1 pack of cigarettes and consumes 250 mL of wine daily. Upper GI endoscopy shows two small ulcers in the body of the stomach. She is diagnosed with peptic ulcer disease because of an increased acid secretion. Blockade of which of the following most likely reduces acid secretion in this patient?

- A. Secretin secretion
- B. Histamine H_2 receptors
- C. Pepsinogen secretion
- D. Somatostatin secretion
- E. CCK secretion

Answer Key: B

Rationale: Histamine H_2 receptors are present on gastric parietal cells and stimulate acid production. Blocking these receptors reduces acid secretion, which is one of the

treatment for acid over-secretion. Secretin mostly acts on bicarbonate release from the pancreas, but it also inhibits gastric acid secretion, hence blocking secretin secretion would enhance gastric acid production. Same thing with somatostatin which inhibits acid secretion. Pepsinogen is not related to acid secretion.

5. A 26-year-old woman comes to the office because of a 2-month history of burning pain below the sternum. He says that the pain begins in the epigastric area one hour after eating and is associated with nausea. He has had some relief from the proton pump inhibitor that was prescribed 2 weeks ago. Physical examination shows no abnormalities. Laboratory studies show an extremely high serum concentration of gastrin. Imaging studies show a mass in the tail of the pancreas. He is diagnosed with a gastrin-secreting tumor. Which of the following mechanisms best explains this patient's symptoms?

- A. Reduced gastric motility
- B. Stimulation of the parietal cells
- C. Lack of blood flow to the stomach
- D. Failure of relaxation of the lower esophageal sphincter
- E. Gall bladder contraction

Answer Key: B

Rationale: This patient has a tumor that secretes hormone, gastrin. Physiologically, gastrin is also secreted from the G cells of the stomach. Gastrin stimulates parietal cells to secrete acid and intrinsic factor.

6. A 49-year-old man comes to the clinic because of a 10-month history of intermittent abdominal pain. Physical examination shows epigastric tenderness to palpation. Gastric juice study shows increased acid secretion. Stool and blood studies show no abnormalities. An upper GI endoscopy shows an ulcer near the gastric antrum. He is diagnosed with peptic ulcer disease and prescribed a drug that inhibits acid secretion from the parietal cells of the stomach. Which of the following mechanisms in the parietal cells is most likely blocked by this drug?

- A. Hydrogen-potassium ATPase
- B. Acetylcholine receptors
- C. Bicarbonate-chloride exchanger
- D. Carbonic anhydrase
- E. Histamine receptors

Answer Key: A

Rationale: Acid secretion occurs through an active transporter called H^+-K^+ ATPase located in the luminal surface of the parietal cells. Chloride moves to the lumen through chloride channel. The source of H^+ is from CO_2 . The bicarbonate produced by this process moves to the blood in exchange with chloride through a bicarbonate-chloride exchanger located in the basal surface of the parietal cell. When acid is secreted into the

stomach, the corresponding HCO_3^- (alkalinity) diffuses via the interstitial space into gastric venous blood. This is also called the 'alkaline tide.'

7. A 38-year-old woman comes to the physician for a follow-up examination. She has a 2-year history of autoimmune disease of the salivary gland. Physical examination shows dry mouth and scanty salivary secretion. Histological examination of a biopsy of the salivary gland shows atrophy of the gland. Which of the following gastrointestinal functions is most likely impaired in this patient?

- A. Digestion of proteins in mouth
- B. Reabsorption of vitamin B12
- C. Carbohydrate digestion in mouth
- D. Carbohydrate digestion in small intestine
- E. Digestion of fat in small intestine

Answer Key: C

Rationale: The saliva contains enzyme, amylase which helps in digestion of carbohydrates in mouth. This enzyme gets inactivated in stomach because of the acidic pH.

8. A 61-year-old man comes to the clinic because of a 4-month history of burning epigastric pain. The pain is relieved with meals. He has no history of nausea or vomiting. An acid secretion stimulation test shows a gastric juice pH of 0.94 (Normal: 1.6 to 3.5). Secretion of which of the following substances most likely inhibits acid secretion in this patient?

- A. Gastrin
- B. Acetylcholine
- C. Histamine
- D. Gastrin-releasing peptide
- E. Secretin

Answer Key: E

Rationale: Acid secretion by parietal cells is stimulated by all above factors except secretin. Secretin and somatostatin inhibit acid secretion.

9. A 32-year-old woman comes to the physician because of an 8-month history of progressive fatigue. In the beginning, she felt more tired than usual at the end of the day but now has difficulty walking up one flight of stairs. She also has numbness and tingling in her feet. Her menses occur at regular 28-day cycle. Her vital signs are within normal limits. Physical examination shows pale tongue and sclera, and ulcers at the angle of the mouth. Sensory examination shows diminished proprioception and vibration sensation in the lower extremities. A peripheral blood smear shows macrocytic hypochromic red blood

cells and hypersegmented neutrophils. Histologic examination of a gastric mucosal biopsy shows loss of gastric glands. She is diagnosed with vitamin B12 deficiency anemia secondary to stomach disorder. Reduced secretion of which of the following substances from the stomach is the most likely cause of this patient's presentation?

- A. Hydrochloric acid
- B. Mucin
- C. Gastrin
- D. Intrinsic factor
- E. Pepsinogen
- F. Lipase

Answer Key: D

Rationale: Parietal cells of the stomach secrete hydrochloric acid and intrinsic factor (IF). IF is essential for absorption of vitamin B12 in the ileum. This patient has loss of gastric glands and therefore has deficient secretion of both HCl and IF. IF deficiency reduced vitamin B12 absorption and anemia. The type of anemia due to vitamin B12 deficiency is called megaloblastic anemia in which the red cells are reduced in number (and hence reduced hemoglobin concentration) but increase in size.

10. A 24-year-old woman comes to the physician for a routine examination. She has no complaints but gets increased salivary secretion for certain food stuff. She loves spicy food from childhood. She says she starts salivating when she smells and sees spicy food. Which of the following mechanisms is responsible for increased salivation in this patient?

- A. Stimulation of the sympathetic nerves to the salivary glands
- B. Increased blood flow to the salivary glands
- C. Conditioned reflex
- D. Stimulation of the enteric nervous system
- E. Vago vagal reflex

Answer Key: C

Rationale: This is a classic example for the conditioned reflex. Thought, smell, or sight of food stimulates salivary nucleus in the brain located in the brain stem. Activation of these centers stimulates salivary glands through the parasympathetic nerves (VII and IX cranial nerves) supplying these glands. This is called cephalic phase of digestion which is mediated through long reflex pathways.